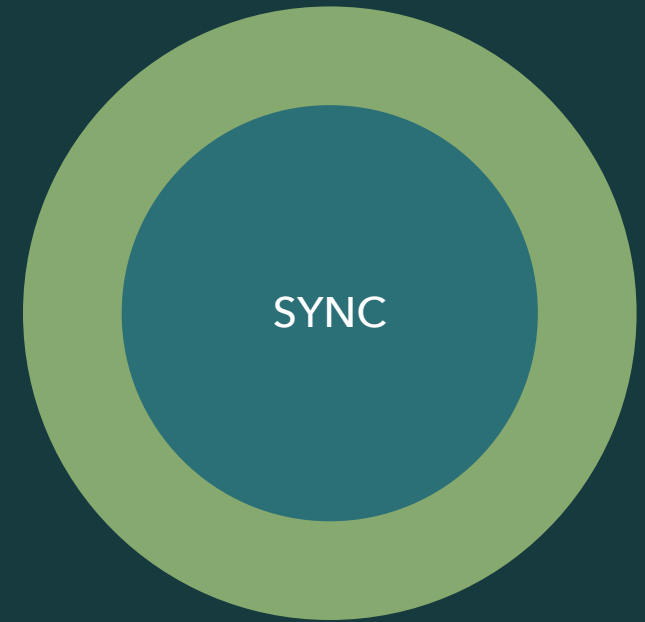


Neuradix POS

Technical Architecture

Local Multi-Shop selective replication, metadata boundaries, encryption, relay routing, and verified release gates.



One client, four persistence models

Commercial caps remain backend configuration, never frontend constants.

FREE LOCAL MULTI-SHOP

Metadata only

Shop-local inventory; peer-synced customers and finalized sales.

SHARED CLOUD STARTER

Shared doctypes

Capped products, customers, sales, and sale lines with business isolation.

SHARED CLOUD PAID

Shared doctypes

Higher or unrestricted configured limits and cloud history.

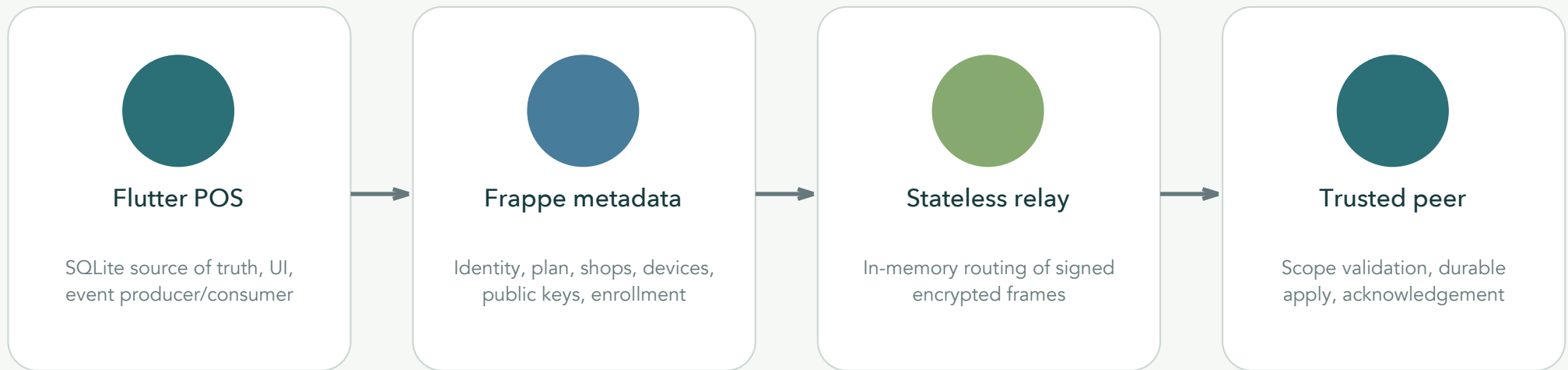
DEDICATED BACKEND

Connector-owned

All business data persists through a connector such as ERPNext. Odoo and Neuradix Atlas Team are planned targets.

Runtime components

Operational plaintext exists only inside trusted clients.



The relay never receives a plaintext payload and has no durable mailbox.

Storage boundary

Local Multi-Shop separates metadata from operational data.

ON TRUSTED DEVICES

- Shop item master, prices, barcodes, images, and stock
- Business customer replicas
- Finalized sales and immutable sale-line snapshots
- Encrypted outbox/inbox, acknowledgements, and conflicts
- Drafts, parked carts, and local settings
- Private keys and decrypted business key

ON NEURADIX INFRASTRUCTURE

- Identity, membership, and subscription state
- Shop and trusted-device metadata
- Device public keys and opaque key envelopes
- Short-lived relay authorization
- Temporary ciphertext in relay memory
- No Product, Customer, Sale, or Sale Item rows

Selective replication contract

Sale-line snapshots travel for history but never become another shop's inventory.

ENTITY	ROUTE	RECIPIENT	ACK SET
Inventory item	Same shop	Online registers assigned to origin shop	Same-shop active peers
Customer	Business-wide	All online trusted business registers	All active business peers
Finalized sale	Business-wide	All online trusted business registers	All active business peers
Draft / parked	None	Current register only	Not applicable

Transactional write and delivery flow

Acknowledgement follows durable apply, never mere transmission.

1 Local commit

Business record and outgoing event share one SQLite transaction.

2 Protect

AES-GCM encrypts payload; Ed25519 signs the envelope.

3 Authorize

Token binds business, device, shop, epoch, and protocol.

4 Route

Relay validates origin shop and selects eligible online peers.

5 Apply

Peer verifies, decrypts, validates scope, and commits atomically.

6 Acknowledge

Origin settles only after required peers confirm durable apply.

Trust and cryptography

Metadata may identify a device; only a trusted device can decrypt business events.

Ed25519

Signs each event and authenticates its origin device.

X25519

Protects the business-key envelope during enrollment.

AES-GCM

Encrypts operational payloads and detects tampering.

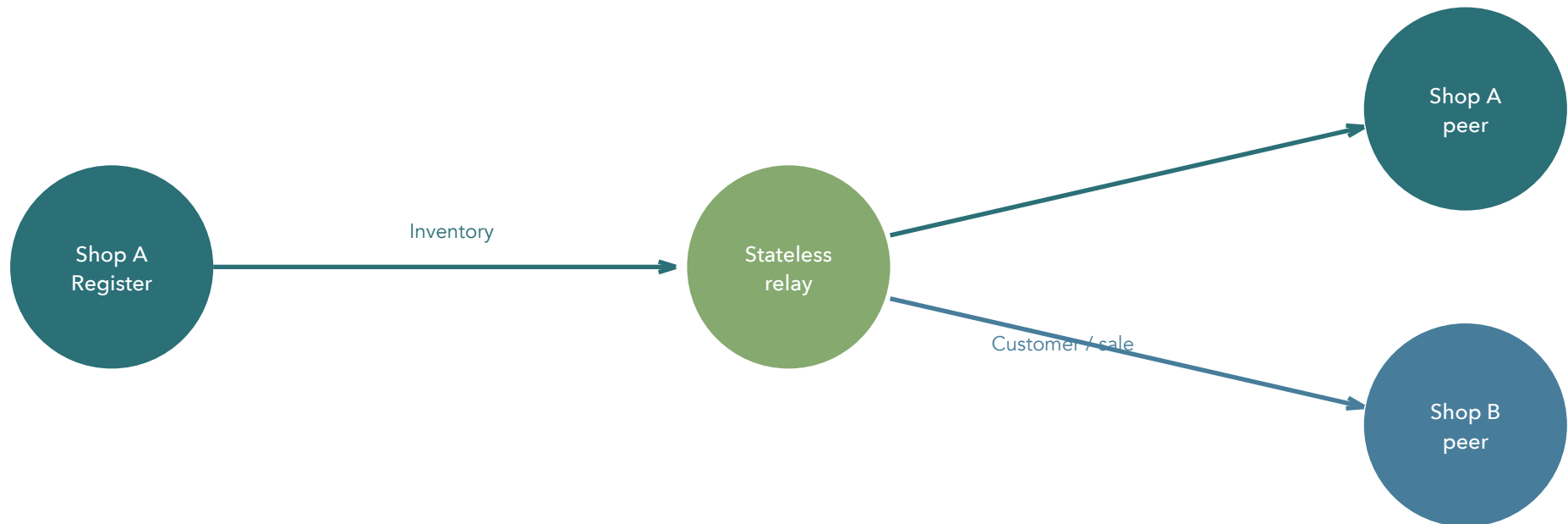
HLC

Orders customer updates without trusting wall clocks alone.

Plaintext business keys and decrypted payloads are never written to Frappe or relay logs.

Relay routing and acknowledgement sets

Recipient scope and required acknowledgements use the same event policy.



Inventory ACK waits only for active Shop A peers. Customer/sale ACK waits for every active business peer.

Registration and enrollment boundary

All flows are available to Flutter; no manual backend seeding is required.

`register_business`

Owner, business, membership, subscription, optional device, first shop

`get_metadata`

Shops, trusted devices, current assignment, pending enrollment, relay URL

`create_shop`

Owner-only unique shop creation

`start_enrollment`

Selected shop, device public keys, expiring request

`approve_enrollment`

Trusted owner wraps business key for the new device

`complete_enrollment`

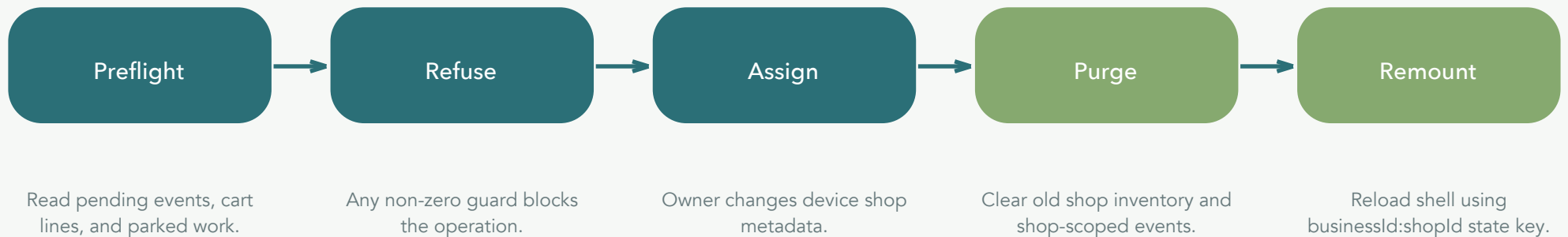
Device accepts envelope and becomes trusted

`issue_relay_token`

Short-lived token including authenticated shop_id

Safe shop reassignment lifecycle

Changing shop scope is refused while local work could be stranded.



Business-wide customers and finalized sales survive reassignment; the prior shop catalog does not.

Server certification result

Two real Android emulators, deployed metadata API, deployed stateless relay.



Release gates and operational limits

The tested scope is green; the production hardening backlog remains explicit.

PASSING GATES

- Backend and connector regressions
- Flutter format, analysis, unit, repository, and widget tests
- Deterministic two-register matrix
- ADB server/emulator matrix
- Raw SQL no-operational-storage audit
- PDF rendering and content validation

PRODUCTION HARDENING

- HTTPS and WSS endpoints
- Relay supervision, monitoring, and revocation runbook
- QR camera pairing and operator PINs
- SQLCipher and managed device backup policy
- Snapshot/anti-entropy recovery
- Explicit warning: total device loss is unrecoverable